



Saurabh Singh

Computer Science and Engineering

Bachelor of Technology

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GitHub Profile

LinkedIn Profile

Portfolio

EDUCATION

St. Andrews Institute of Technology and Management (SAITM)

2022 – 2026

Bachelor of Technology in Computer Science and Engineering (AI & ML)

CGPA: 8.0

EXPERIENCE

Zanze Company

Jan 2026 – Mar 2026

AI Engineer Intern

Remote

- Architected and deployed an **AI-powered Chatbot** using **LangChain** and **OpenAI GPT API**, enabling intelligent, context-aware conversational responses for real-world enterprise business queries.
- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline with **FAISS vector store** and semantic embeddings, improving response accuracy by grounding chatbot answers in company-specific knowledge bases.
- Integrated the chatbot into a production-ready **Streamlit** web application, delivering a seamless user experience with multi-turn conversational memory and context-aware reply generation.

PROJECTS

Real-Time Face Detection System | TensorFlow, CNN, VGG16, OpenCV, Matplotlib

Source Code

- Designed and trained a **CNN-based Face Detection System** leveraging **VGG16 transfer learning** with TensorFlow/Keras, achieving high detection accuracy using the Adam optimizer and backpropagation.
- Implemented robust data preprocessing, normalization, and augmentation pipelines to enhance model generalization; integrated **OpenCV** for real-time webcam detection with bounding box visualization.
- Monitored and evaluated model performance using accuracy and loss metrics visualized through **Matplotlib**, ensuring reliable and production-grade face detection results.

Chat With PDF | Python, LangChain, FAISS, RAG, Streamlit

Source Code

- Built an intelligent **Generative AI document assistant** using LLMs, LangChain, and OpenAI API to deliver precise, context-aware question answering over user-uploaded PDF documents.
- Implemented a full **RAG pipeline** with PDF chunking (PyPDF2), **FAISS vector database**, and semantic embeddings for accurate and efficient information retrieval at scale.
- Designed an interactive **Streamlit** front-end with multi-turn conversational memory, enabling real-time document

TECHNICAL SKILLS

Programming Languages: Python (Advanced), C++ (Advanced), SQL, R

Machine Learning & Deep Learning: CNN, RNN, Encoder-Decoder, Transformer Architecture, Backpropagation, Optimization Techniques

Generative AI & LLMs: LangChain, RAG, Prompt Engineering, AI Agents, FAISS, VD

Frameworks & Libraries: TensorFlow, NumPy, Pandas, PyTorch (Basics), OpenCV, Flask, Streamlit, PyPDF2

Backend & Deployment: REST APIs, Git, GitHub, Docker, Jupyter Notebook

DevOps & Tools: VS Code,

Relevant Coursework: DAA, OOPS, DBMS, CN, OS (Windows, Linux)

Languages: English (Professional), Hindi (Native)

ACHIEVEMENTS

- Solved **200+ coding challenges** on LeetCode, CodeForce, and CodeChef platforms.
- Secured **2nd Rank** in College Science Exhibition.

CERTIFICATIONS

- Data Science, Machine Learning, DL, NLP Bootcamp** – Udemy (2025)
- Full Stack Generative and Agentic AI with Python** – Udemy (2025)
- Data Science Course** – PW Skills (2024)
- AI Learning** – IBM Skillbuild (2023)
- UI/UX Design** – Udemy (2022)